

Teacher:	Ray Joshua Balingkit	**Subject:**	Design & Technology
Week:	11 (Term 1)	**Year:**	10–11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will explain how a programmable component follows stored instructions to control a system. They will develop their understanding through a teacher explanation, a worked example and an evidence-based classroom task. In the second lesson, students will model a control algorithm with a flowchart and verify it using test cases. Their recorded reasoning, practical evidence and final checks will show whether they can apply the ideas to a defined user, product or design problem.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher questioning, labelled diagrams, case comparisons and the completed independent tasks. Check accurate terminology, evidence linked to the user or brief and a reasoned explanation of choices. Review recorded tests or traces alongside conclusions, and use the final check in both lessons to identify the next skill needing practice.

LESSON SLOTS - TUESDAY AND WEDNESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	3 November 2026	By the end of this lesson, students will be able to explain how a programmable component follows stored instructions to control a system.	**H:** Compare a fixed switch with a controller that can follow different instructions. Ask what must change if a designer wants a different response without replacing the whole product. **ID:** Use the programmable components section to introduce a microcontroller as a component that runs stored instructions, reads inputs and controls outputs. Distinguish the hardware, program and external components. Explain sequence, selection and repetition through familiar control examples. **WD:** Trace a teacher-prepared controller that reads a button and switches an indicator on while pressed. Identify the input value, decision, output and repeated check. Show that changing the decision rule	**Support:** Provide hardware-program-input-output cards and a step-by-step trace table. Physically act out one cycle before students trace another input independently. **Extension:** Students identify a case where changing the program is insufficient because a different sensor or output device is needed, and justify that distinction.

			changes behaviour while the diagram of the main components can remain the same. **YD:** Students label a controller diagram and trace two short instruction sequences against supplied input values. They predict outputs, identify a missing repeated check and explain the difference between changing hardware and changing the program. **C:** Ask students why instructions must be unambiguous and why checking an input only once may be inadequate. Collect a trace to check that the predicted output follows the stored rule.	
2	4 Novembe r 2026	By the end of this lesson, students will be able to model a control algorithm with a flowchart and verify it using test cases.	**H:** Display a flowchart whose decision has only a Yes path. Ask what happens when the input does not meet the condition and how that omission affects the product. **ID:** Introduce start/end, process, input/output and decision symbols and the use of labelled arrows. Explain that a flowchart must communicate the intended sequence and all possible decision outcomes. A loop repeats checks; the model still needs defined behaviour when a condition is false. **WD:** Draw a flowchart for the dark-room lamp rule from the previous week. Trace below, equal-to and above-threshold inputs, including a change from dark to bright. Model correcting a missing off instruction and show how a retest confirms the correction. **YD:** Students construct a	**Support:** Provide symbol cards, an incomplete chart and a test table. Build the first decision together and use arrows that can be moved before committing to a final drawing. **Extension:** Students add an explicit second condition and devise a test that distinguishes the revised algorithm from the original, explaining the extra complexity.

			control flowchart for a supplied sensor-and-output brief. They trace at least four cases, including a boundary and a changing input, record expected and observed model outputs and correct one inconsistency. Use paper traces or an available classroom simulation; save the finished chart and evidence. **C:** Ask a partner to follow the chart without verbal help. Check decision labels, complete paths and whether the repeated check allows a changed input to update the output.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel GCSE (9–1) Design and Technology revision notes: 1.7 The use of programmable components
- Teacher computer, display screen and whiteboard markers
- Prepared product scenarios, diagrams and user briefs aligned to the lesson objectives
- Prepared analysis worksheet, comparison grid or test-recording table
- Paper, pencils, rulers, card templates and approved classroom stationery
- Teacher-prepared samples or component cards; pre-cut model parts where required
- Lesson success criteria, observation checklist and labelled evidence folders



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?